T2523 addendum — the atlas re-reading, the full cross-check sweep, and the Šilov-asymptotic scope

This is NOT a new theorem. The confinement result is T2523 (2026-07-23, Tier D), banked four weeks ago; I cite it and register only the delta. Count once. My "T2572" is retracted as a double-count (counter released back to 2572). The re-derivation shed T2523's load-bearing scope; this note restores it and adds three genuinely new things. New standing rule this earned: when you re-derive, grep before registering — not for priority, but to inherit the scope.

The theorem (T2523, verbatim scope — restored)

> T2523 (2026-07-23, Tier D): "Colored ⇔ $\lambda_2 > 0$ ⇔ Šilov-vanishing." A K-type with λ_2>0 has zero boundary value (K744) → cannot reach the boundary → confined; λ_2=0 reaches it → free.

The load-bearing (A)/(B) split (marked "external-safety" in T2523; every "confinement is derived" claim MUST carry it): - (A) DERIVED: *no free colored asymptotic states* — a colored (λ_2>0) mode has identically zero Šilov support, so it cannot be an asymptotic state. This is the "no free quarks or gluons" notion. - (B) NOT DERIVED: the center / area-law / string-tension / mass-gap confinement — the Yang–Mills Millennium notion. BST does not derive (B) via T2523. *(Consistent with the YM arc: the boundary gap was a KK artifact, not the Clay gap, K1714.)*

My re-derivation stated "confinement = the kernel of the Šilov projection" with no (A)/(B) split — exactly the unscoped form T2523 exists to forbid. Corrected: the reading is (A)-only.

The criterion is $\lambda_2 > 0$, NOT N-ality (T2523, better than I had it): the adjoint gluon has N-ality 0 yet $\lambda_2 > 0$, so it is (A)-confined (no free gluon). An "N-ality $\neq 0$" phrasing would wrongly free the gluon and break the theorem. Keep the criterion λ_2>0.

Delta 1 — the atlas re-reading (count-once with T2523)

In the atlas's transition-map language, T2523's confinement is the statement that the Šilov (boundary-value) projection Π_Š has kernel = the colored sector (λ_2>0). This is a *re-reading* of T2523 in the dictionary's verbs, not a new result — it places (A)-confinement as the load-bearing example of a transition-with-a-kernel (the atlas's first physics reading). No new theorem; T2523 is cited.

Delta 2 — the cross-check sweep, WITH its denominator (C6)

3/3 with an unstated denominator is what C6 forbids. Here is the full list — every state with a clean K-type assignment and a known confinement status — and its outcome:

state	K-type / λ_2	predicted	observed	outcome
charged lepton	color-singlet, λ_2=0	emitted (free)	free	[yes]
neutrino	color-singlet, λ_2=0	emitted (free)	free	[yes]
photon / U(1)_em	color-singlet, λ_2=0	free	free	[yes]
W / Z	color-singlet, λ_2=0	free (asymptotic)	free	[yes]
Higgs (1,0)	color-singlet, λ_2=0	boundary-reaching/observable	observable	[yes]
quark (triplet)	λ_2>0	(A)-confined (no free quark)	no free quark	[yes]
adjoint gluon	λ_2>0, N-ality 0	(A)-confined (no free gluon)	no free gluon	[yes] (the sharp test)
glueball	color-singlet composite, λ_2=0	emittable in principle	(searched)	[yes] (consistent)

8 checks, 8 consistent, 0 counterexamples. The denominator is stated; the sharp test is the adjoint gluon (λ_2>0, N-ality 0), which T2523's criterion gets right where an N-ality criterion would fail. No known state with a clean K-type contradicts (A). *(This cross-validation is the genuine addition — it is not in T2523.)*

Delta 3 — is the Šilov boundary the physical asymptotic region, or a compactification artifact? (pre-empt §621)

The first question from any referee who read §621 (which killed the YM gap by finding *compactness* under a number). The (A)-confinement mechanism is vanishing on the compact Šilov boundary — so the question is fair: does "λ_2>0 modes vanish at the boundary" survive decompactification, or is it a statement about modes on a finite drum?

The honest pre-emption — and why I do *not* expect the §621 kill here: unlike the YM gap (whose *value* scaled $1/a^2 \to 0$ under decompactification, revealing a KK artifact), asymptotics is what a boundary is *for*. For the Hardy space H^2(D_IV^5), a holomorphic function's boundary values *are* its asymptotic data — the Šilov boundary is the natural asymptotic region of the domain by construction, not an incidental compactification. So the vanishing of colored boundary support is a statement about asymptotic states in the domain's *own* asymptotics, and there is no radius a to send to infinity: the boundary is load-bearing, not a regulator.

What stays open (Cal's to sharpen): whether D_IV^5's Šilov boundary is the physical spacetime asymptotic region — versus an *internal* boundary — is part of the geometry–physics seam (K1716). (A)-confinement holds in the domain's own asymptotics; its identification with physical asymptotic infinity is the seam, and is not claimed settled here. So the theorem is safe against the §621 attack on its own terms (no radius to decompactify), while the seam question is flagged, not swept.

Ledger / registry disposition

- T2523 stands (Tier D); count unmoved. No new theorem registered. T2572 retracted (double-count); counter released.
- The three deltas (atlas re-reading, cross-check sweep, asymptotic paragraph) are recorded here as a T2523 addendum, scoped to (A).
- Standing rules earned: (1) re-derivation inherits its scope by grep; (2) sweep provenance after a fix, not just before (Elie).

Lyra, 2026-08-21 (T2523 addendum). RETRACTED my T2572 as a double-count of T2523 (2026-07-23, Tier D, "Colored $\Leftrightarrow$ $\lambda_2>0$ $\Leftrightarrow$ Šilov-vanishing"); counter released to 2572; count once. RESTORED the load-bearing scope: (A) no free colored asymptotic states DERIVED, (B) area-law/string-tension/mass-gap NOT derived; criterion is $\lambda_2>0$ (adjoint gluon (A)-confined), not N-ality. DELTAS: (1) atlas re-reading = ker Π_Š, count-once; (2) full cross-check sweep 8/8 with denominator stated (C6), sharp test = adjoint gluon; (3) Šilov-asymptotic paragraph — no §621 kill (asymptotics is what a boundary is for; no radius to decompactify), physical-asymptotic identification flagged as the open seam (K1716, Cal's to sharpen). Nothing pushed; CP existence-only.